

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation is strictly maintained across all Python files.
2	Proper File Structure	Files and folders are logically organized	Yes	Clean separation with folders for uploads/, templates/, and root files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names used (e.g., transcribed_text, duration_seconds).
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly state their intent (e.g., evaluate_concept_with_gemini, save_waveform_image).
5	Code Comments	Inline and block comments explain logic	Yes	Comments added for all key modules like Database Setup, AI Evaluation, and PDF Generation.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separated core logic into independent helper functions instead of single large blocks.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reused common parameters and avoided hardcoding redundant values.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Implemented robust try-except blocks with proper safety fallbacks for API and DB calls.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Database Initialization & Connection	Python / SQLite3	Used during server startup, audio uploading, and fetching history records.	High
2	Gemini AI Evaluation Logic	Python / Google GenAI SDK	Reusable for any academic text or voice analysis module in the future.	High
3	Audio Waveform Generator	Python / Matplotlib & Wave	Can be reused in any frontend dashboard or report generation section to plot audio data.	Medium
4	ReportLab PDF Document Engine	Python / ReportLab	Reused across the application whenever a student scorecard or report needs to be generated.	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The code follows standard PEP 8 layout guidelines with an organized structure.
Readability	5	Variables and functions are clearly named, making it easy for any developer to understand.
Reusability	4	Most logic is modularized into independent functions, reducing dependencies.
Documentation / Comments	4	Well-placed inline comments and section dividers document the codebase effectively.
Overall Score	4.5	The application code is clean, production-ready, maintainable, and highly reusable.